

Verification Continuum™

**AEware Application Note
for
compute_fta_all**



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AE-ware Application Note for – compute_fta_all

This document describes the purpose and usage of the “compute_fta_all” AE-ware.

For product support, see <https://www.synopsys.com/support.html>.

Introduction

When performing FTA analysis, the user would like to consider all FPV proven properties as well as all the inconclusives FPV properties at their respective safe depths.

Currently, the in-built tool commands either allows the user to consider only the proven properties or proven+inconclusive at a min depth (the min depth gets applied to proven also, which the user would not want). In order to work around this limitation, the user needs to code a set of tcl commands to achieve a more comprehensive FTA analysis.

This AE-ware provides this capability. This command allows the user to perform comprehensive FTA analysis by using the proven properties first and then using the inconclusives properties at their respective depths.

This command also allows the user to bucket the inconclusive properties into unique depth buckets for faster runtime.

Prerequisites

- 1) VC Formal version needs to be 2021.09-1 or higher.
- 2) The availability of FPV results, that is, the FPV run should have been performed before moving onto formal core analysis.

Location

`$VC_STATIC_HOME/auxx/ctg/tcl/library/proc/compute_fta_all.tbc`

Command Help

`compute_fta_all -help`

Usage: `compute_fta_all` # Compute FTA for both proven and inconclusive properties from FPV run. For inconclusive properties, analysis is done based on unique safe depth buckets or per property bases

`[-max_time_proven ]` (Max time to be spent on proven FPV properties (Default: 1H))

`[-max_time_inconcl ]` (Max time to be spent on inconcl FPV properties (Default: 1H))

`[-mode_for_inconcl ]` (Values: 0 (Default. Unique safe depth buckets), 1 (Each property). Only applicable for inconclusives.)

`[-resume]` (Skip Fault activation and do a continuation run)

Usage

The following steps need to be followed to use this AE-ware :

1. Run `check_fv` for FPV.
2. Source the AEware script.
3. Execute “`compute_fta_all`” command.

Example

```
vcf>.....
```

```
#run FPV
```

```
vcf> check_fv
```

```
#source AEware
```

```
vcf>Source
```

```
$::env($VC_STATIC_HOME)/auxx/ctg/tcl/library/proc/compute_fta_all.tbc
```

```
vcf>compute_fta_all -max_time_proven 30M -max_time_inconcl 5M -  
mode_for_inconcl 0
```

(In the above example we are setting max time for proven as 30 minutes and for each inconclusive unique safe depth bucket as 5 minutes)

Sample AE-ware output

```
386 vcf>source compute_fta_all.tbc
387 vcf>compute_fta_all -max_time_proven 30M -max_time_inconcl 5M -mode_for_inconcl 0
388 [AEware] Starting Comprehensive FTA Analysis
389 [AEware] WARNING : Liveness properties are not supported in FTA yet
390 [AEware] Identifying Inconclusive Property Depths
...
508 [AEware] The unique safe depths for FTA are : 9 10 11 12 14 15 16 17 20 29 68
509 [AEware] Pre-processing complete
510 [AEware] Starting step-by-step FTA run
511 [AEware] STEP-1 : Fault activation analysis
512 [Info] FTA_ACTIVATE_START: Activation analysis ongoing, cannot stop now, please wait.
...
536 [AEware] STEP-1 : Fault activation complete
537 [AEware] STEP-2 : FTA run with proven FPV properties started
...
933 [AEware] STEP-2 : FTA run with proven FPV properties complete
934 [AEware] STEP-3 : FTA run with inconclusive FPV properties started
935 [AEware] Mode 0 selected : Running based on unique depth buckets
936 [AEware] Bucket 0 Unique depth : 9
```

Known Limitations

Currently Liveness properties are not supported for FTA analysis, hence only the safety properties are filtered in this AE-ware.

References

Please refer to *Section 19.7 VC Formal User Guide* for more info on formal core analysis in VC Formal.